

Things To Know About RF POWER MEASUREMENT

EFY BUREAU

This article talks about the essential characteristics, types and performance factors of power sensors and power meters in RF power measurement



Radio frequency (RF) and microwave power measurements are important for measuring the performance of a device under test (DUT). Selection of measuring instruments and techniques is based on type of measurement. Understanding the different types of RF power meters, and the ways in which errors in RF and microwave power measurements can be minimised are key in making accurate RF power readings.

Power sensors and spectrum analysers are two key instruments for RF power measurements. Selection of equipment is done on the basis of its dynamic range, accuracy, frequency range, durability and portability.

Thermal or diode-based detectors are used for RF power sensing. The complete setup of a power sensor and a power meter consists of a sensor element on the front, followed by amplifier, filter, analogue-to-digital converter (ADC), microprocessor and display unit. Incoming RF signal is converted to DC or low-frequency voltage waveform by this sensor element. Waveform is then amplified and filtered.

Some setups have increased filter

bandwidth to improve measurement speed, or reduced to improve measurement sensitivity. An ADC converts analogue signals into digital ones. A microprocessor handles additional filtering and time averaging of the sampled waveform, as well as interfacing to the display.

Effects of uncertainties on RF power measurement

Despite their features, equipment for RF power measurement are sometimes subjected to different factors that may affect the quality of results. Some examples of sources of uncertainty are mismatches, linearity errors, interference and temperature effects. To evaluate total uncertainty, it is important to determine and combine all sources of uncertainty to obtain the combined uncertainty.

Generally, accurate RF power measurement is difficult to do as frequency is high (1GHz to 100GHz), apart from complex modulation types and formats. Highest measurement accuracy for absolute power can be obtained through a power sensor. Error sources of the sensor include linearity

uncertainty, zero offset and drift, calibration uncertainty, temperature effects and display noise.

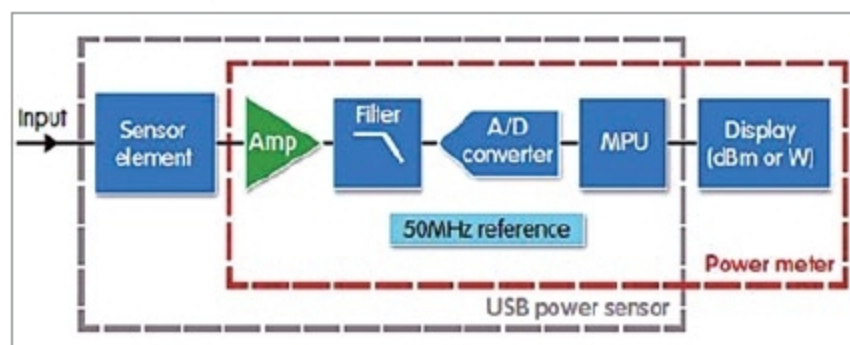
These sources of errors should be taken into account while RF power measurement. For absolute power

measurement, uncertainty in reference measurement with the power sensor is the main source of error. Two crucial factors affecting this are sensor specification and voltage standing-wave ratio (VSWR) occurring due to a mismatch between the sensor and source.

John Hamilton, vice president - marketing, RFMW Ltd, says "The higher the VSWR, the greater the uncertainty! Metrology-grade components have very low VSWR, which minimises unwanted reflections. A minimum of 10dB signal-to-noise ratio (SNR) is necessary to make accurate measurements with these devices. Inherent errors can be minimised through a calibration process. It is essential to calibrate on a regular basis, since temperature, test cable flexure and even connection quality affect the calibration of test equipment."

Types of RF power meters

Different types of RF power meters are used under different circumstances. Absorptive RF power meter is the most widely-used for accurate laboratory measurements, while through-line power meter is used mostly for field applications. Hamilton adds, "Continuous wave is the most common. It provides an average power measurement that is highly accurate and traceable. With certain applications, such as radar, pulsed RF measurements are required and provide pulse power information. With modulation (FM, AM, QAM, etc), special considerations are required to provide instantaneous



Block diagram of power sensor and power meter (Credit: New Electronics)